



NEET 2.0 2027

Physics

Lecture - 7

Units and Measurement

By- Manish Raj (MR Sir)



Today's Goal

Questions on error

- ✓ Date → 27/7/2026
- H/w → goal

→ Udm Sangharsh
Assignment - 3 → P12

→ Udm MR. Physics
upto error

→ [Maha Mantra of
Udm]

Error

$$\Delta x = \text{Absolute error} = |x_T - x_m|$$

- have unit same as P-Q
- can't tell about accuracy of precision
- always +

unitless
dim⁰ ✓

$$\left[\begin{array}{l} \text{relative error} = \frac{\Delta x}{x_T} \\ \text{error} = \frac{\Delta x}{x_T} \times 100 \end{array} \right]$$

How to write error

$$(x \pm \Delta x) \text{ unit}$$

$$(x \pm x\%)$$

$$y = A + B$$
$$\Delta y = \underline{\Delta A} + \underline{\Delta B} \checkmark$$

$$\frac{\Delta y}{y} = \frac{\Delta A + \Delta B}{A + B} \checkmark$$

$$y = A - B$$
$$\Delta y = \underline{\Delta A} + \underline{\Delta B}$$

$$\frac{\Delta y}{y} = \frac{\Delta A + \Delta B}{A - B} \checkmark$$

$$y = A \cdot B \quad \Delta y = A \Delta B + B \Delta A \quad y = A/B$$

$$\Delta y = \Delta A \cdot \Delta B \quad \times$$

$$\frac{\Delta y}{y} = \left(\frac{\Delta A}{A} \right) + \left(\frac{\Delta B}{B} \right) \checkmark$$

$$\frac{\Delta y}{y} = \frac{\Delta A}{A} + \frac{\Delta B}{B} \checkmark$$

$$y = A^m B^p$$

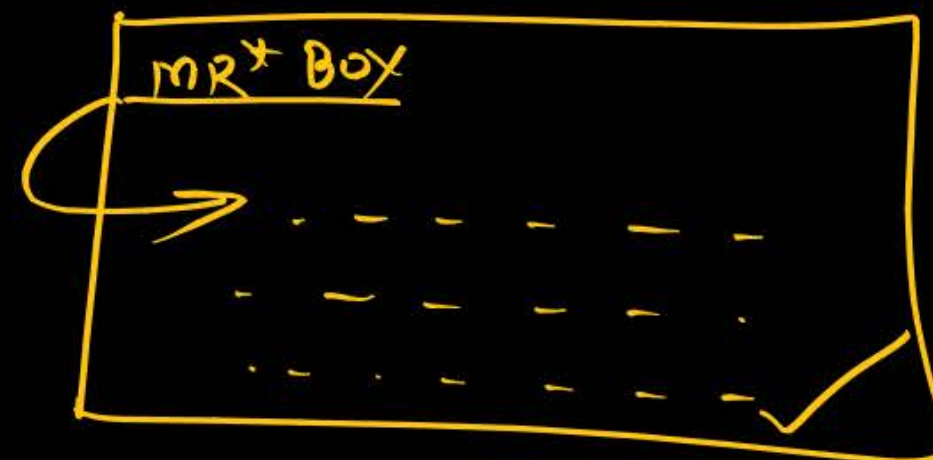
$$\frac{\Delta y}{y} = m \frac{\Delta A}{A} + p \frac{\Delta B}{B}$$

Copy in not2

MR* Box

Mathematical expression me +ve ya
-ve the write absolute error
directly.

If division, multiplication or Power
then write relative
error directly.



Question

No copy



A physical quantity is represented by $X = [M^a L^b T^{-c}]$. If percentage error in the measurement of M, L and T are $\alpha\%$, $\beta\%$ and $\gamma\%$ respectively, then maximum percentage error in measurement of X should be (Given that α , β and γ are very small)

NEET/TEJ

~~1~~ $(\alpha a - \beta b + \gamma c)\%$

$$X = \frac{M^a L^b}{T^c} \checkmark$$

2 $(\alpha a + \beta b + \gamma c)\%$ ✓

~~3~~ $(\alpha a - \beta b - \gamma c)\%$

~~4~~ $(\alpha a + \beta b - \gamma c)\%$

Question

copy



A physical quantity P is given as

$$P = \frac{a^2 b^3}{c \sqrt{d}}$$

The percentage error in the measurement of a , b , c and d are 1%, 2%, 3% and 4% respectively. The percentage error in the measurement of quantity P will be

[10 Apr, 2023]

1 13%

2 14%

3 12%

4 16%

$$P = \frac{a^2 b^3}{c \sqrt{d}}$$

$$\begin{aligned} 100 \times \frac{\Delta P}{P} &= \left[2 \frac{\Delta a}{a} + 3 \frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{1}{2} \frac{\Delta d}{d} \right] \times 100 \\ &= 2 \times 1 + 3 \times 2 + 3 + \frac{1}{2} \times 4 = 13\% \end{aligned}$$

Question

no copy



A rectangular plate has length (2 ± 0.02) cm and width (1 ± 0.01) cm. The maximum percentage error in the measurement of its area is

1 1%

2 2%

3 3%

4 5%

$$l = (2 \pm 0.02) \text{ cm}$$

↑ ↓
length Δl

$$d = (1 \pm 0.01)$$

↑ ↓
width Δd

$$A = l \times d$$

$$\frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta d}{d}$$

$$100 \times \frac{\Delta A}{A} = \left[\frac{0.02}{2} + \frac{0.01}{1} \right] \times 100$$

$$= 0.02 \times 100$$

$$= 2\%$$

Question

Copy (solve in rough)
Now



A body of mass (5 ± 0.5) kg is moving with a velocity of (20 ± 0.4) m/s. Its kinetic energy will be
[13 Apr, 2023]

1 ~~(1000 $\pm$ 140) J~~
 26%
 $\Delta K.E$

2 ~~(1000 $\pm$ 0.14) J~~
 62%
 $\Delta K.E$

3 (500 $\pm$ 0.14) J

4 (500 $\pm$ 140) J

$$K.E = \left(\frac{1}{2}\right)mv^2 = \frac{1}{2}mv^2 = 1000J$$

$$\left(\frac{\Delta K.E}{K.E}\right) = \frac{\Delta m}{m} + 2 \frac{\Delta v}{v}$$

$$\frac{\Delta K.E}{K.E} = \frac{0.5}{5} + \frac{2 \times 0.4}{20}$$

$$= 0.1 + 0.04$$

$$\frac{\Delta K.E}{K.E} = 0.14$$

$$\therefore \text{Error} = \frac{\Delta K.E}{K.E} \times 100 = 0.14 \times 100 = 14\%$$

$$\Delta K.E = K.E \times 0.14$$

$$= \frac{1}{2} \times 5 \times (20)^2 \times 0.14$$

$$= \frac{1}{2} \times 5 \times 400 \times 0.14$$

$$= 1000 \times 0.14$$

$$\Delta K.E = 140$$

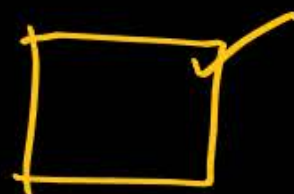
$$K.E = 1000 \checkmark$$

Question

Copy



A public park, in the form of a square, has an area of $(100 \pm 0.2) \text{ m}^2$. The side of park is ✓



$$\rightarrow \text{Area} = 100 \text{ m}^2$$
$$\Delta A = 0.2 \text{ m}^2$$

1 $(10 \pm 0.01) \text{ m}$ (571)

absolute

2 $(10 \pm 0.1) \text{ m}$ ✗

3 $(10 \pm 0.02) \text{ m}$ ✗

4 $(10 \pm 0.2) \text{ m}$ ✗

$$\Delta l = \frac{l}{2} \frac{\Delta A}{A}$$
$$= \frac{10}{2} \times \frac{0.2}{100}$$

$$= \frac{0.1}{10}$$

$$= 0.01$$

$$A = l^2$$

$$l = \sqrt{A} = \sqrt{100}$$
$$l = 10 \text{ m}$$

$$\rightarrow l = A^{1/2}$$

$$\left[\frac{\Delta l}{l} \right] = \frac{1}{2} \frac{\Delta A}{A}$$

Mr* Box

Sabse Pahle
p.q calculate kre ✓
then usme
absolute or %
calculate kre ✓

Question

No Log



A force F is applied on a square area of side L . If the percentage error in the measurement of L is 2% and that in F is 4%, what is the maximum percentage error in pressure?

1 2%

2 4%

3 6%

4 8%

$$\frac{\Delta L}{L} \times 100 = 2\% \quad \checkmark$$

$$\frac{\Delta F}{F} \times 100 = 4\% \quad \checkmark$$

$$P = \frac{F}{A} = \frac{F}{L^2}$$

$$\frac{\Delta P}{P} = \left(\frac{\Delta F}{F} \right) + 2 \frac{\Delta L}{L}$$

Question

copy



A cylindrical wire of mass $(0.4 \pm 0.01)\text{g}$ has length $(8 \pm 0.04)\text{cm}$ and radius $(6 \pm 0.03)\text{mm}$. The maximum error in its density will be **[08 Apr, 2023]** ✓

- 1 1%
- 2 3.5%
- 3 4% ✓
- 4 5%



$$V = \pi r^2 l$$

$$\begin{cases} m = 0.4 \text{ g} \\ \Delta m = 0.01 \text{ g} \end{cases}$$

$$\begin{cases} l = 8 \text{ cm} \\ \Delta l = 0.04 \text{ cm} \end{cases}$$

$$\begin{cases} r = 6 \text{ mm} \\ \Delta r = 0.03 \text{ mm} \end{cases}$$

$$\rho = \frac{m}{\pi r^2 l}$$

$$100 \times \frac{\Delta \rho}{\rho} = \left[\frac{\Delta m}{m} + 2 \frac{\Delta r}{r} + \frac{\Delta l}{l} \right] \times 100$$

$$= \frac{0.01}{0.4} \times 100 + \left[2 \frac{0.03}{6} \times 100 \right] + \frac{0.04}{8} \times 100$$

$$= \frac{1}{0.4} + 1 + 0.5$$

$$= \frac{10}{4} + 1.5 = 4\%$$

Question

No copy



If $Z = \frac{A^2 B^3}{C^4}$, then the relative error in Z will be:

[25 June, 2022]

1 $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$

2 $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$

3 $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$ ✓

4 $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

Question

H/O NO COPY



A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy

[27 July, 2022]

$$\tau = \frac{m^1}{L^2} \frac{1}{T^2}$$

1 15%

2 25% ✓

3 75%

4 5%

Question

No copy



The acceleration due to gravity is measured on the surface of earth by using a simple pendulum. If α and β are relative errors in the measurement of length and time period respectively, then percentage error in the measurement of acceleration due to gravity is

1 $(\alpha + \frac{1}{2}\beta)$ ✗

2 $(\alpha - 2\beta)$ ✗

3 $(2\alpha + \beta) \times 100$ ✗

4 $(\alpha + 2\beta) \times 100$ ✓

hint $T = 2\pi \sqrt{\frac{l}{g}}$

$$T^2 = \frac{l}{g}$$

$$g = \frac{l}{T^2}$$

$$\left[100 \frac{\Delta g}{g} \right] = \left(\frac{\Delta l}{l} \right) + 2 \frac{\Delta T}{T}$$

$$\frac{\Delta g}{g} \times 100 = (\alpha + 2\beta) \times 100$$

Question

$$\underline{10\text{ mm} = 1\text{ cm}} \quad 1\text{ mm} = 0.1\text{ cm}$$

$$\begin{aligned} \text{total time} \rightarrow t &= nT \\ \left(\frac{\Delta t}{t}\right) &= \frac{\Delta T}{T} \end{aligned}$$



The time period of a simple pendulum is given by $T = 2\pi \sqrt{\frac{l}{g}}$. The measured value of the length of pendulum is 10 cm known to a 1 mm accuracy. The time for 200 oscillations of the pendulum is found to be 100 second using a clock of 1s resolution. The percentage accuracy in the determination of 'g' using this pendulum is 'x'. The value of 'x' to the nearest integer is:

1 0.03 ✓✓

2 0.05

3 0.04

4 0.02

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$T^2 = \frac{l}{g}$$

$$g = \frac{l}{T^2}$$

$$\begin{aligned} \frac{\Delta g}{g} \times 100 &= \left(\frac{\Delta l}{l} + 2 \frac{\Delta T}{T} \right) \times 100 \\ &= \left\{ \frac{0.1\text{ cm}}{10\text{ cm}} + 2 \left[\frac{1\text{ s}}{100} \right] \right\} \times 100 \\ &= 1 + 2 \left[\frac{1}{100} \times 100 \right] \\ &= 1 + 2 = 3 \end{aligned}$$

Question

Copy



$y = \sin \theta$ find percentage error in y if percentage error in θ is 2% at $\theta = \pi/6$ rad.

Soln

" "

$$y = \sin \theta$$

diffⁿ w.r.t θ

$$\frac{dy}{d\theta} = \frac{d(\sin \theta)}{d\theta}$$

$$\frac{dy}{d\theta} = \cos \theta$$

$$dy = \cos \theta d\theta$$

divid by y both sides

$$\frac{dy}{y} = \frac{\cos \theta d\theta}{\sin \theta}$$

given

$$\frac{\Delta \theta}{\theta} \times 100 = 2$$

$$\theta = \pi/6$$

$$\frac{\Delta y}{y} \times 100 = ??$$

$$100 \times \frac{\Delta y}{y} = \cot \theta \boxed{\Delta \theta \times 100}$$

% error in y

$$= \cot \theta \cdot 2\theta$$

$$= \cot(\pi/6) \times 2 \times \frac{\pi}{6}$$

$$= \sqrt{3} \times \frac{\pi}{3} = \left(\frac{\pi}{\sqrt{3}} \right) \%$$

$$\boxed{\cot(\pi/6) = \sqrt{3}}$$

Question

Copy $\sin(180) = 0$



If $y = m \tan \theta$, where m is constant then find angle at which percentage error in 'y' will be minimum.

$$\theta = ??$$

$$\left(\frac{dy}{y} \times 100 \right) \text{ minimum}$$

$$y = m \tan \theta$$

diff'n w.r.t θ

$$\frac{dy}{d\theta} = m \sec^2 \theta$$

$$dy = \frac{m d\theta}{\cos^2 \theta}$$

divide by y both side

$$\frac{dy}{y} = \frac{m d\theta}{\cos^2 \theta \times m \frac{\sin \theta}{\cos \theta}}$$

$$\left(\frac{dy}{y} \right) = \frac{d\theta}{\cos \theta \cdot \sin \theta}$$

$$\left(100 \frac{dy}{y} \right)_{\text{minimum}} = \frac{2 d\theta \times 100}{[\sin(2\theta)]_{\text{max}}}$$

$$\sin(2\theta) = 2 \sin \theta \cdot \cos \theta$$

$$\sin(2\theta) = 1$$

$$\text{if } 2\theta = 90$$

$$\theta = 45^\circ$$

Ans

Gpy in not 2
Q1 Error in the measurement of Resistance in Series Combination

$$\text{gf } R_1 = (10 \pm 1) \Omega$$
$$R_2 = (20 \pm 1) \Omega$$

Find % error in R

(v) $R = R_1 R_2$

$$100 \times \frac{\Delta R}{R} = \left(\frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} \right) \times 100$$

$$= \left(\frac{1}{10} + \frac{1}{20} \right) \times 100$$

$$= 10 + 5 = 15\%$$

(vi) find absolute error in $R = R_1 R_2$

$$= 10 \times 20$$

(iv) find error in R

where $R = R_2 - R_1$

$$R = R_2 - R_1 = 20 - 10 = 10 \Omega$$

$$\Delta R = \Delta R_2 + \Delta R_1 = 1 + 1 = 2$$

$$\frac{\Delta R}{R} = \frac{2}{10} \times 100 = \underline{\underline{20\%}}$$

(i) find % error in R_1

$$\frac{\Delta R_1}{R_1} \times 100 = \frac{1}{10} \times 100 = 10\%$$

(ii) find % error in R_2

$$\frac{\Delta R_2}{R_2} \times 100 = \frac{1}{20} \times 100 = 5\%$$

(iii) find % error in R in series comb.

$$R = R_1 + R_2 = 30 \Omega$$

$$\Delta R = \Delta R_1 + \Delta R_2 = 1 + 1 = 2$$

$$\frac{\Delta R}{R} \times 100 = \frac{2}{30} \times 100 = \frac{20}{3} = 6.6\%$$

$$\frac{\Delta R}{R} = \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2}$$

$$\Delta R = R \left[\frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} \right]$$

$$\Delta R = 200 \left[\frac{1}{10} + \frac{1}{20} \right]$$

$$= 200 \left(\frac{2+1}{20} \right)$$

$$= \frac{200 \times 3}{20}$$

$$\boxed{\Delta R = 30} \Omega$$

(iv) find absolute in R when

$$R = \frac{R_2}{R_1}$$

$$R = \frac{20}{10} = 2$$

$$\frac{\Delta R}{R} = \frac{\Delta R_2}{R_2} + \frac{\Delta R_1}{R_1}$$

$$\Delta R = R \left[\frac{\Delta R_2}{R_2} + \frac{\Delta R_1}{R_1} \right]$$

$$= 2 \left[\frac{1}{20} + \frac{1}{10} \right]$$

$$\boxed{\Delta R = 0.3 R}$$

$$\frac{\Delta R}{R} \times 100 = \frac{0.3}{2} \times 100 = 15\%$$

$$= 2 \left[\frac{1+2}{10} \right] = \frac{3}{5} = 0.3$$

copy

$$y = x^2 \longrightarrow \frac{dy}{dx} = 2x^{2-1} = 2x$$

$$y = x^3 \longrightarrow \frac{dy}{dx} = 3x^{3-1} = 3x^2$$

$$y = \frac{1}{x^2} = x^{-2} \longrightarrow \frac{dy}{dx} = -2x^{-2-1} = -2x^{-3} = \frac{-2}{x^3}$$

$$y = \frac{1}{x} \longrightarrow \frac{dy}{dx} = \frac{dx^{-1}}{dx} = -1x^{-2} = \frac{-1}{x^2}$$

$y = \frac{1}{x}$

$$\frac{dy}{dt} = -\frac{1}{x^2} \left(\frac{dx}{dt} \right)$$

→ 10:21 to
10:26
tax

5 min one
leg stand

error in the measurement of Resistance in Parallel combination

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

diffⁿ w.r.t 't' both side

$$-\frac{1}{R_{eq}^2} \frac{\Delta R_{eq}}{\Delta t} = -\frac{1}{R_1^2} \frac{\Delta R_1}{\Delta t} - \frac{1}{R_2^2} \frac{\Delta R_2}{\Delta t}$$

$$\left(\frac{\Delta R_{eq}}{R_{eq}^2} \right) = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\Delta R_{eq} (\text{absolute error}) = R_{eq}^2 \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2} \right)$$

$$\frac{\Delta R_{eq}}{R_{eq}} = R_{eq} \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2} \right)$$

MR SCAM

Kis Type Ka error hai

(a) relative

(b) absolute

(c) % error

(d) Kuch Nahi

18% (+1)

Series combination of capacitor

$$\frac{1}{C_g} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{\Delta C_g}{C_g^2} = \frac{\Delta C_1}{C_1^2} + \frac{\Delta C_2}{C_2^2}$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{\Delta f}{f^2} = \frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}$$

Question



Two resistance $R_1 = (20\Omega \pm 2\Omega)$ and $R_2 = (5 \pm \underline{1}\Omega)$ are connected in parallel then find % and absolute error.

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{20 \times 5}{20 + 5} \\ = \frac{100}{25} = 4$$

$$\frac{\Delta R_{eq}}{R_{eq}} \times 100 = R_{eq} \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2} \right) \times 100$$

$$\therefore \text{error in } R_{eq} = 4 \left[\frac{1}{25} + \frac{2}{400} \right] \times 100$$

$$= 4 \left[\frac{16}{25} + \frac{5}{100} \right]$$

$$= 16 + \frac{4}{2} = 18\%$$

Question



In an experiment for determination of the focal length of a thin convex lens, the distance of the object from the lens is 10 ± 0.1 cm and the distance of its real image from the lens is 20 ± 0.2 cm. The error in the determination of focal length of the lens is $n\%$. The value of n is _____.

JEE Advance - 2023

Solⁿ

$$u = 10 \text{ cm}$$
$$\Delta u = 0.1 \text{ cm}$$

$$v = 20$$
$$\Delta v = 0.2$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$f = \frac{10 \times 20}{30} = \frac{20}{3}$$

$$100 \times \frac{\Delta f}{f} = f \left[\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2} \right] \times 100$$

$$\frac{2}{3} + \frac{5 \times 0.2}{3}$$
$$= \frac{2}{3} + \frac{1}{3} = \frac{3}{3} = 1$$
$$= \frac{20}{3} \left[0.1 + \frac{0.2}{4} \right]$$

Question



Using the expression $2d \sin \theta = \lambda$, one calculates the values of d by measuring the corresponding angles θ in the range 0 to 90° . The wavelength λ is exactly known and the error in θ is constant for all values of θ . As θ increases from 0° :

[JEE-Advance 2013]

1 the absolute error in d remains constant

2 the absolute error in d increases

3 the fractional error in d remains constant

4 the fractional error in d decreases
(Relative)

log se kal
kareng
ye method bhi sikho

$$2d \sin \theta = \lambda$$

$$d = \frac{\lambda}{2 \sin \theta} = \frac{\lambda}{2 \sin \theta} \left[\frac{u}{v} \right]$$

diff'n w.r.t. θ using Rule

$$\frac{\partial d}{\partial \theta} = \frac{0 \times \sin \theta - (\cos \theta) \lambda}{(\sin^2 \theta)}$$

$$\frac{\partial d}{d} = - \frac{\cos \theta \times \partial \theta \times \lambda}{\sin^2 \theta \times \lambda} = - \cot \theta \frac{\partial \theta}{\theta}$$

Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by measuring a dimensionless quantity a . If the error in the measurement of a is Δa ($\Delta a/a \ll 1$), then what is the error Δr in determining r ? **(JEE Adv. 2018)**

1 $\frac{\Delta a}{(1+a)^2}$

2 $\frac{-2\Delta a}{(1+a)^2}$

3 $\frac{2\Delta a}{(1-a)^2}$

4 $\frac{2a\Delta a}{(1-a^2)}$

$$r = \frac{1-a}{1+a}$$

diff'n w.r.t 'a'
 $\frac{dr}{da}$ ✓

Question



The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2\text{s}^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of $E(t)$ at $t = 5\text{s}$ is

(JEE Adv. 2015)

$$E = A^2 \times e^{-\alpha t}$$

Kal clu mo
di

The current voltage relation of diode is given by $I = (e^{1000 V/T} - 1)$ mA, where the applied voltage V is in volt and the temperature T is in kelvin. If a student makes an error measuring ± 0.01 V while measuring the current of 5 mA at 300 K, what will be the error in the value of current in mA? **(JEE Adv. 2018)**

- 1 0.2 mA
- 2 0.02 mA
- 3 0.5 mA
- 4 0.05 mA

Question



In an experiment to determine the acceleration due to gravity g the formula used for the time period of a periodic motion is $T = 2\pi \sqrt{\frac{7(R-r)}{5g}}$. The values of R and r are measured to be (60 ± 1) mm and (10 ± 1) mm, respectively. In five successive measurements, the time period is found to be 0.52 s, 0.56 s, 0.57 s, 0.54 s and 0.59 s. The least count of the watch used for the measurement of time period is 0.01 s. Which of the following statement(s) is (are) true? **[JEE-Advance 2016]**

- 1 The error in the measurement of r is 10%
- 2 The error in the measurement of T is 3.57%
- 3 The error in the measurement of T is 2%
- 4 The error in the determined value of g is 11%

Question



Measured mass of object is 2.5 gram find absolute and percentage error in measurement of mass. **[IIT 2006]**

Thank

You